Web Dev Mastery

📘 Topics Covered:

1. CSS Variables
2. Float
3. Z-Index
4. Box-Shadow

🔹 1. CSS Variables

✅ When to Use:

- To **store and reuse** values like colors, spacing, or fonts.
- Helps in **maintaining consistency** and making **updates easier**.

🧠 Explanation:

CSS Variables (also called custom properties) are declared using `--name` and accessed using `var(--name)`. Global variables should be declared in `:root`.

📝 HTML (`index.html`)

```
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="variables.css">
</head>
<body>
  <h1>Welcome</h1>
  <button class="btn">Click Me</button>
</body>
</html>
```

🎨 CSS (`variables.css`)

```
:root {
  --primary-color: #3498db;
  --text-color: white;
  --padding: 12px 24px;
  --border-radius: 6px;
}

body {
  background-color: var(--primary-color);
  color: var(--text-color);
  font-family: sans-serif;
  text-align: center;
  padding-top: 50px;
}

.btn {
  background-color: var(--text-color);
  color: var(--primary-color);
  padding: var(--padding);
  border: none;
  border-radius: var(--border-radius);
  cursor: pointer;
}
```

🔹 2. Float

✅ When to Use:

- For **text wrapping** around images or side-by-side content like **old layouts**.
- Avoid for layout control; use Flexbox or Grid instead.

🧠 Explanation:

The `float` property lets elements align left or right, allowing inline content (like text) to flow beside them.

📝 HTML (`float.html`)

```
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="float.css">
</head>
<body>
  <div class="wrapper">
    <img src="https://via.placeholder.com/150" alt="example image">
    <p>This paragraph wraps around the floated image. Float is useful for image-text layouts.</p>
  </div>
</body>
</html>
```

🎨 CSS (`float.css`)

```
.wrapper {
  width: 80%;
  margin: 30px auto;
  overflow: auto;
}

img {
  float: left;
  margin-right: 15px;
  border-radius: 8px;
}

p {
  font-size: 16px;
  line-height: 1.6;
}
```

🔹 3. Z-Index

✅ When to Use:

- To **control the stacking order** of overlapping elements (modals, dropdowns, etc.).
- Must be used with a `position` value like `absolute`, `relative`, or `fixed`.

🧠 Explanation:

Elements with a higher `z-index` appear on top of others with a lower value.

📝 HTML (`zindex.html`)

```
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="zindex.css">
</head>
<body>
  <div class="box red"></div>
  <div class="box blue"></div>
</body>
</html>
```

🎨 CSS (`zindex.css`)

```
.box {
  width: 150px;
  height: 150px;
  position: absolute;
}

.red {
  background: red;
  top: 60px;
  left: 60px;
  z-index: 1;
}

.blue {
  background: blue;
  top: 100px;
  left: 100px;
  z-index: 2;
}
```

🔹 4. Box-Shadow

✅ When to Use:

- To create **depth or elevation** in UI design (buttons, cards, modals).
- Improves visual hierarchy.

🧠 Explanation:

The `box-shadow` property adds a shadow to the outer frame of an element.

css
```
box-shadow: offset-x offset-y blur-radius spread-radius color;
```

📝 HTML (`boxshadow.html`)

html
```
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="boxshadow.css">
</head>
<body>
  <div class="card">
    <h2>Card with Shadow</h2>
    <p>This card has a soft shadow around it to make it pop!</p>
  </div>
</body>
</html>
```

🎨 CSS (`boxshadow.css`)

css
```
body {
  background: #f4f4f4;
  font-family: Arial, sans-serif;
  text-align: center;
  padding-top: 100px;
```

```
}

.card {
  background: white;
  width: 300px;
  margin: auto;
  padding: 20px;
  box-shadow: 0 6px 10px rgba(0, 0, 0, 0.15);
  border-radius: 8px;
}
```

✅ Summary

Topic	Use Case Example	Best For
CSS Variables	Reusable theme values	Consistent design
Float	Image/text alignment	Simple inline wrapping
Z-Index	Modal over background	Layering UI elements
Box-Shadow	Elevate cards/buttons	Visual enhancement